

Overview of Machine Learning Algorithms: When to Use and When Not to Use

This document provides a concise overview of various machine learning algorithms, detailing their appropriate use cases and scenarios where they may not be the best choice. Understanding these algorithms is crucial for selecting the right approach for specific data problems.

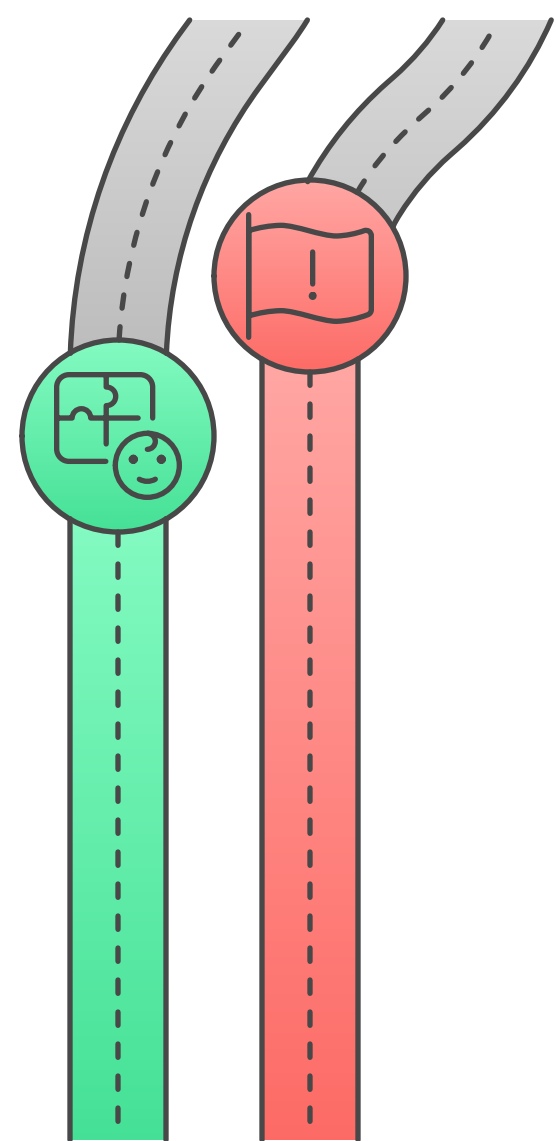
How to choose the right machine learning algorithm?

Use Case Fit

Selecting an algorithm that aligns well with the specific data problem can lead to better outcomes.

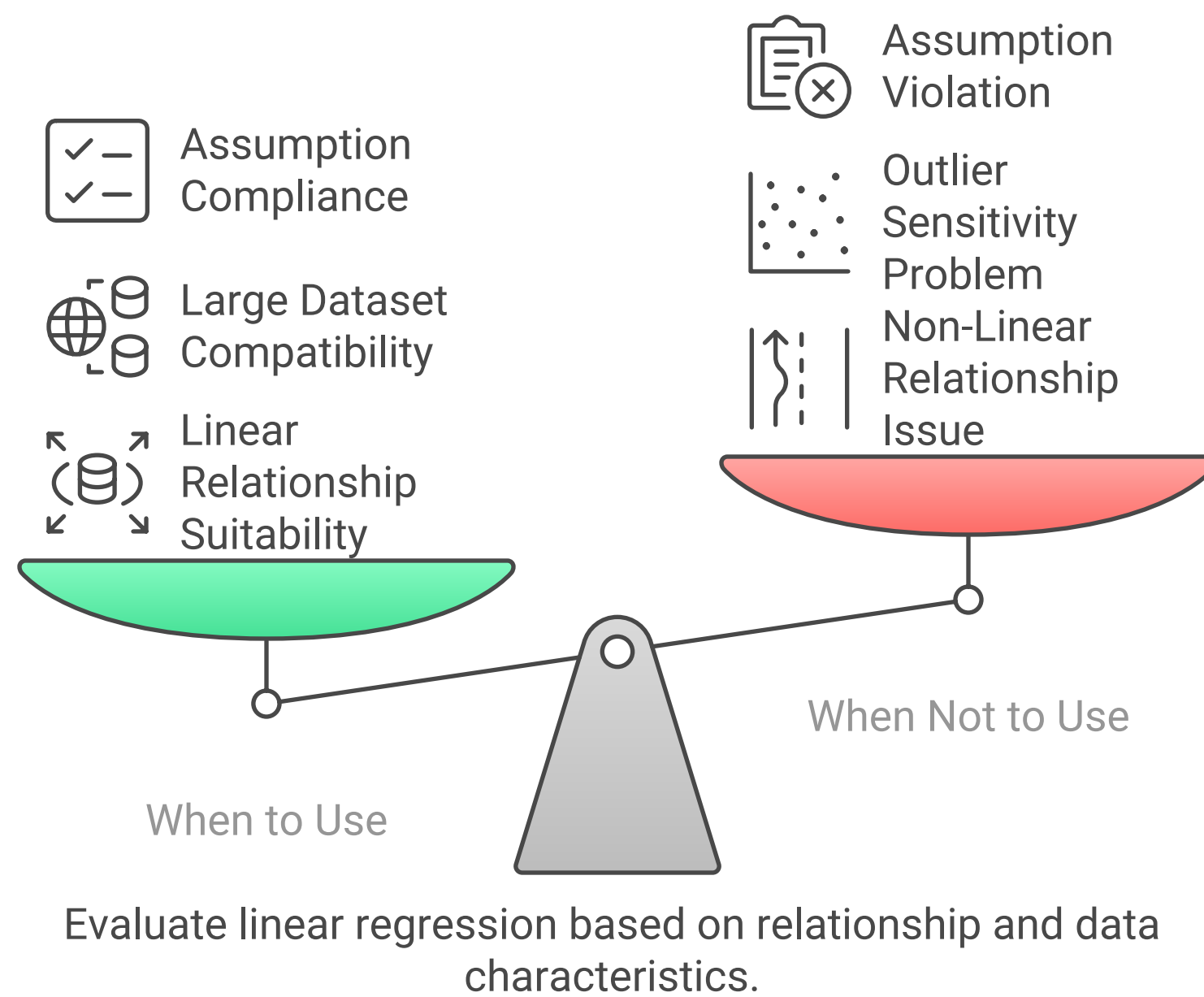
Not Ideal Scenarios

Avoiding algorithms that are unsuitable for the data problem prevents potential pitfalls.



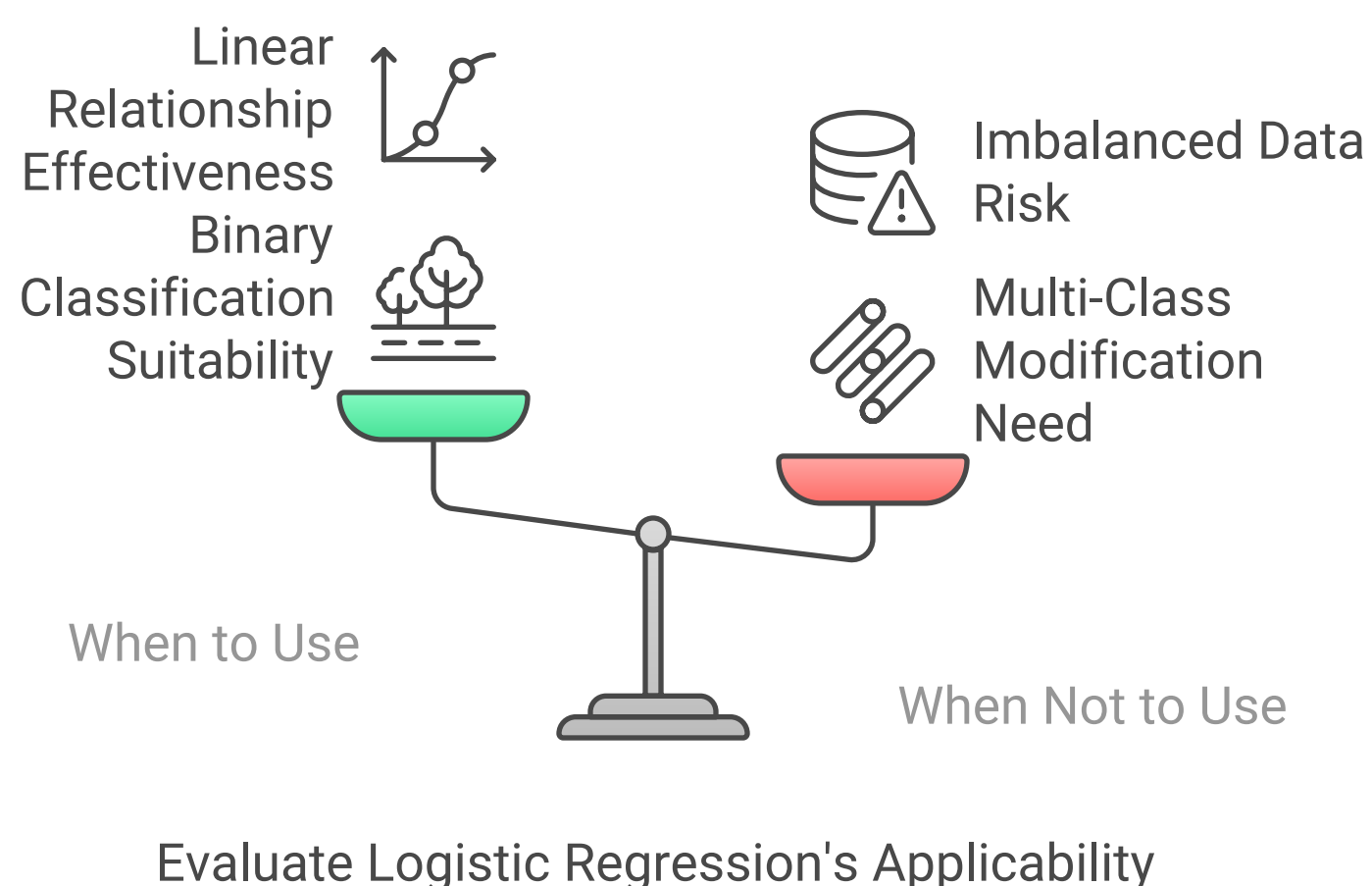
Linear Regression

- **When to Use:** Linear regression is ideal for predicting continuous outcomes when there is a linear relationship between the independent and dependent variables. It works well with large datasets and when the assumptions of linearity, normality, and homoscedasticity are met.
- **When Not to Use:** Avoid using linear regression when the relationship between variables is non-linear, when there are outliers that can skew results, or when the data does not meet the assumptions of the model.



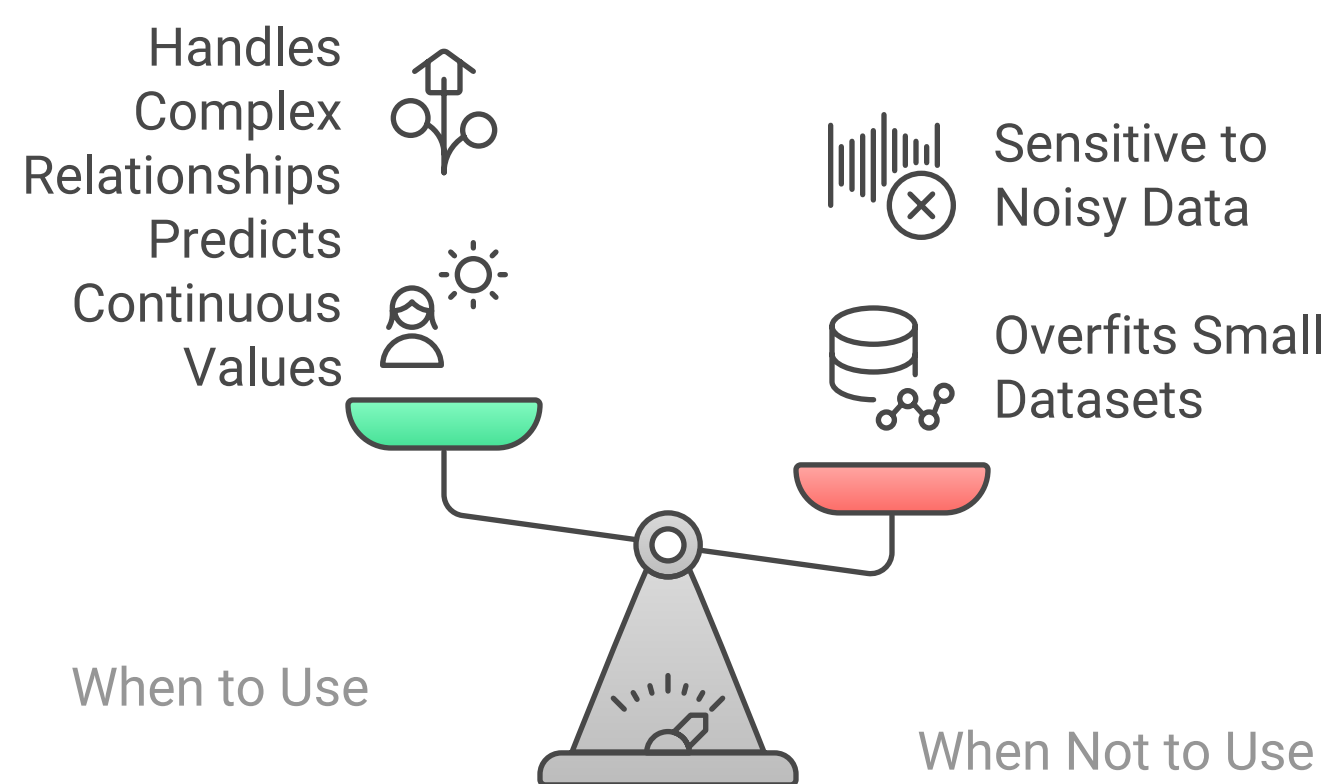
Logistic Regression

- **When to Use:** Logistic regression is suitable for binary classification problems where the outcome is categorical. It is effective when the relationship between the independent variables and the log-odds of the dependent variable is linear.
- **When Not to Use:** It should not be used for multi-class classification without modification (e.g., one-vs-all), or when the data is highly imbalanced, as it may lead to biased predictions.



Decision Tree Regressor

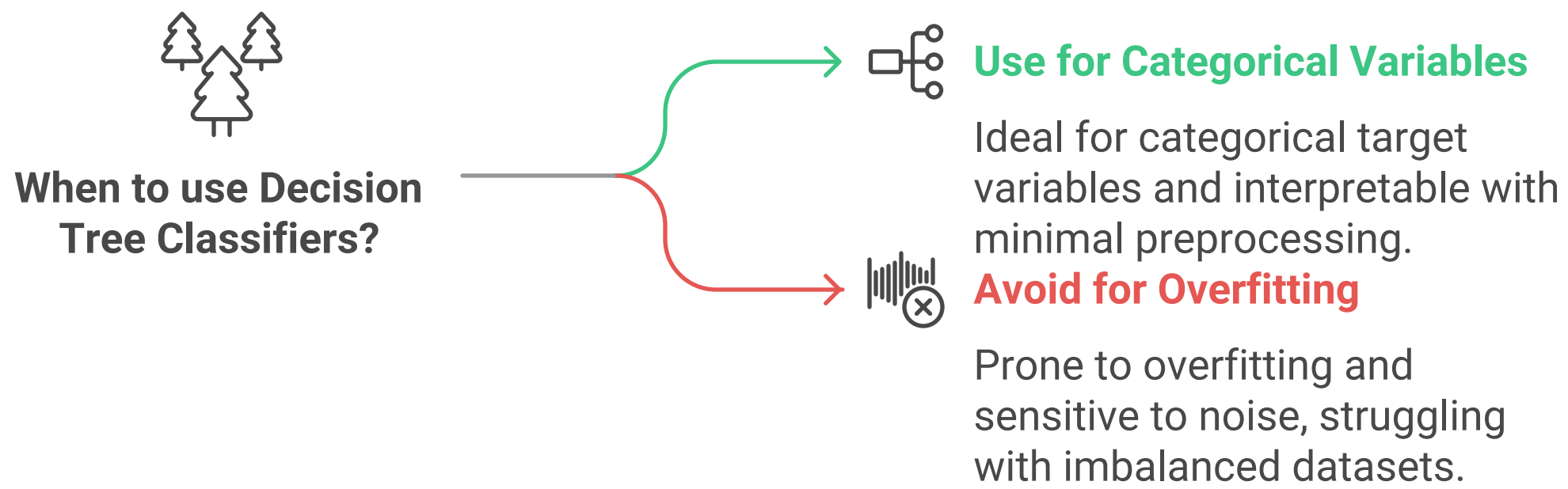
- **When to Use:** This algorithm is useful for predicting continuous values and can handle both numerical and categorical data. It is particularly effective when the relationship between features and the target variable is complex and non-linear.
- **When Not to Use:** Decision trees can overfit the training data, especially with small datasets, and are sensitive to noisy data. They may not perform well in high-dimensional spaces without proper pruning.



Balancing Decision Tree Strengths and Weaknesses

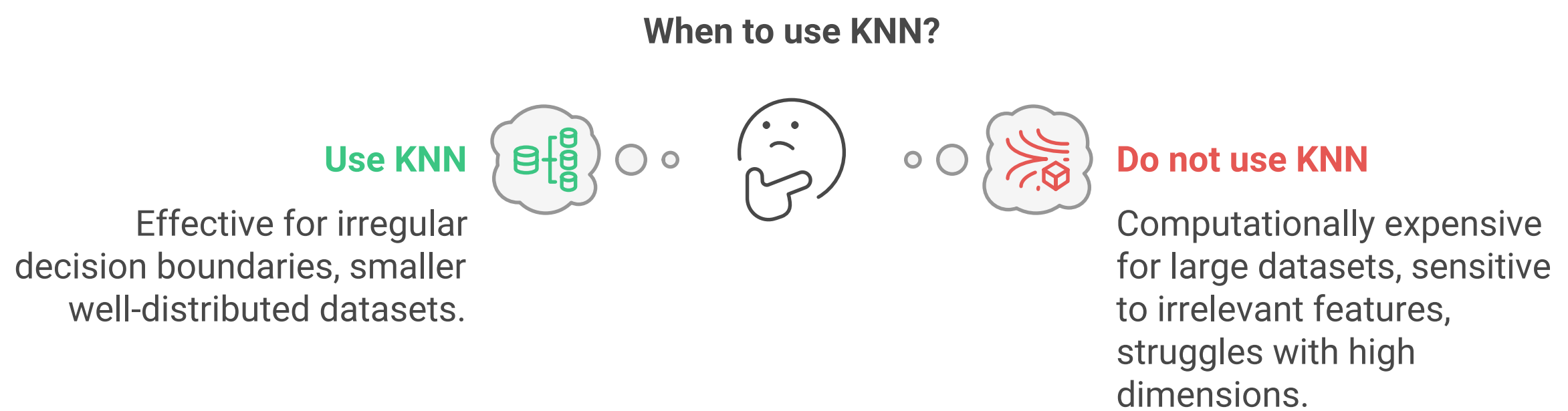
Decision Tree Classifier

- **When to Use:** Decision tree classifiers are great for categorical target variables and can handle both binary and multi-class classification tasks. They are interpretable and require little data preprocessing.
- **When Not to Use:** Similar to the regressor, they can overfit and are sensitive to noise. They may also struggle with imbalanced datasets.



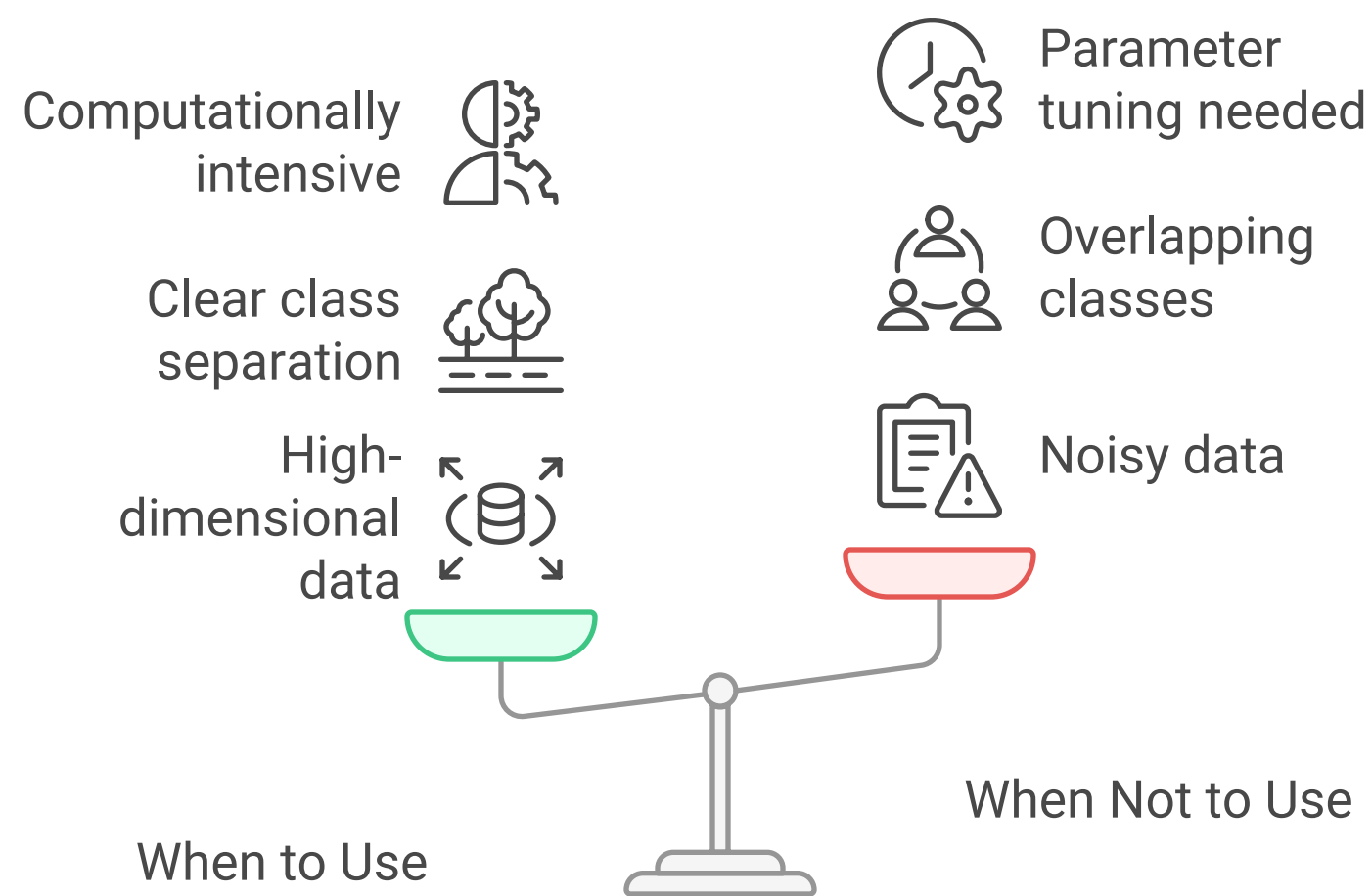
K-Nearest Neighbors (KNN)

- **When to Use:** KNN is effective for both classification and regression tasks, particularly when the decision boundary is irregular. It works well with smaller datasets and when the data is well-distributed.
- **When Not to Use:** KNN can be computationally expensive with large datasets, and it is sensitive to irrelevant features and the choice of distance metric. It also struggles with high-dimensional data due to the curse of dimensionality.



Support Vector Machine (SVM) Classifier

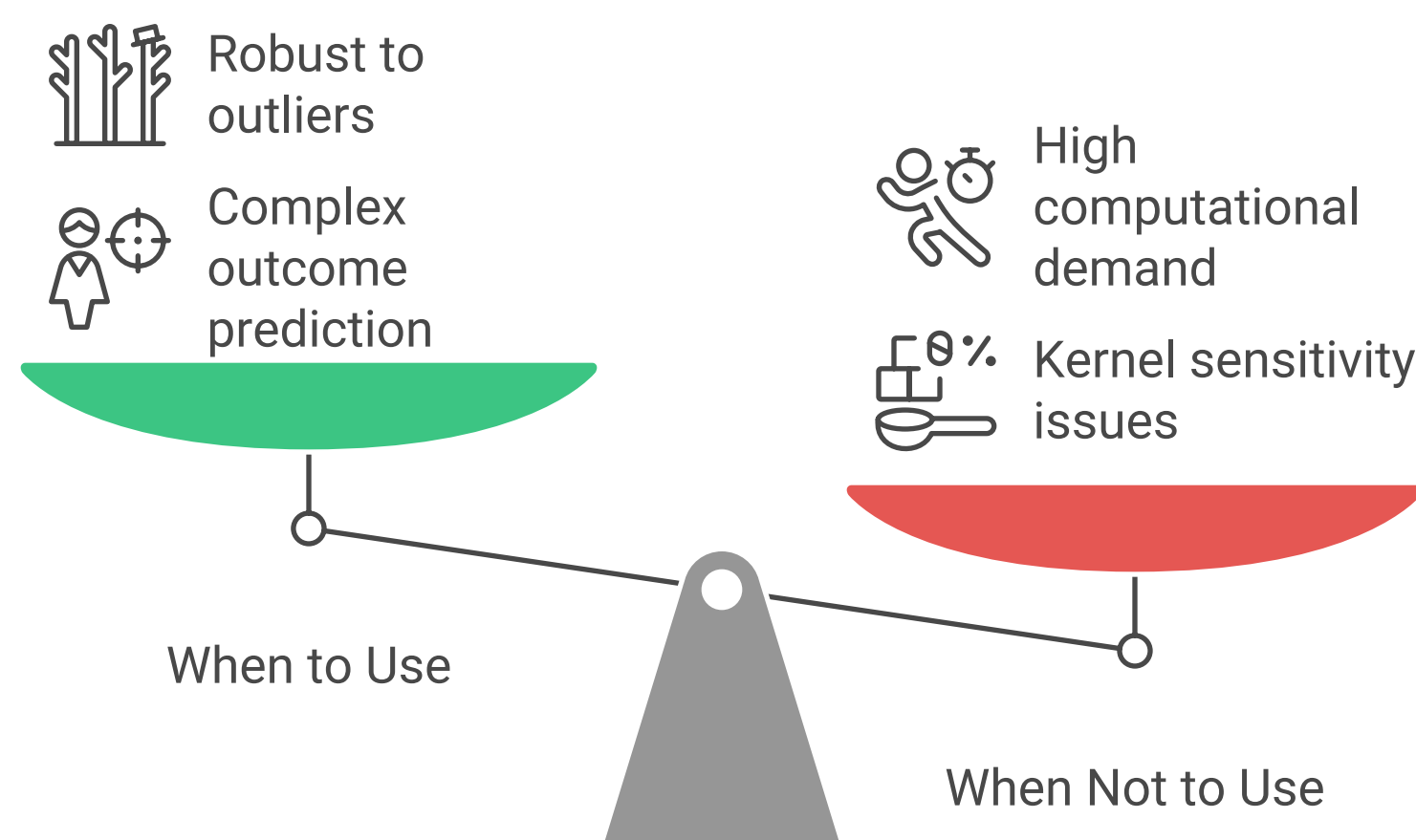
- **When to Use:** SVM classifiers are powerful for binary classification problems, especially with high-dimensional data. They work well when there is a clear margin of separation between classes.
- **When Not to Use:** SVMs can be less effective with noisy data and overlapping classes. They also require careful tuning of parameters and can be computationally intensive.



Evaluate SVM's strengths and weaknesses for optimal use.

Support Vector Machine (SVM) Regressor

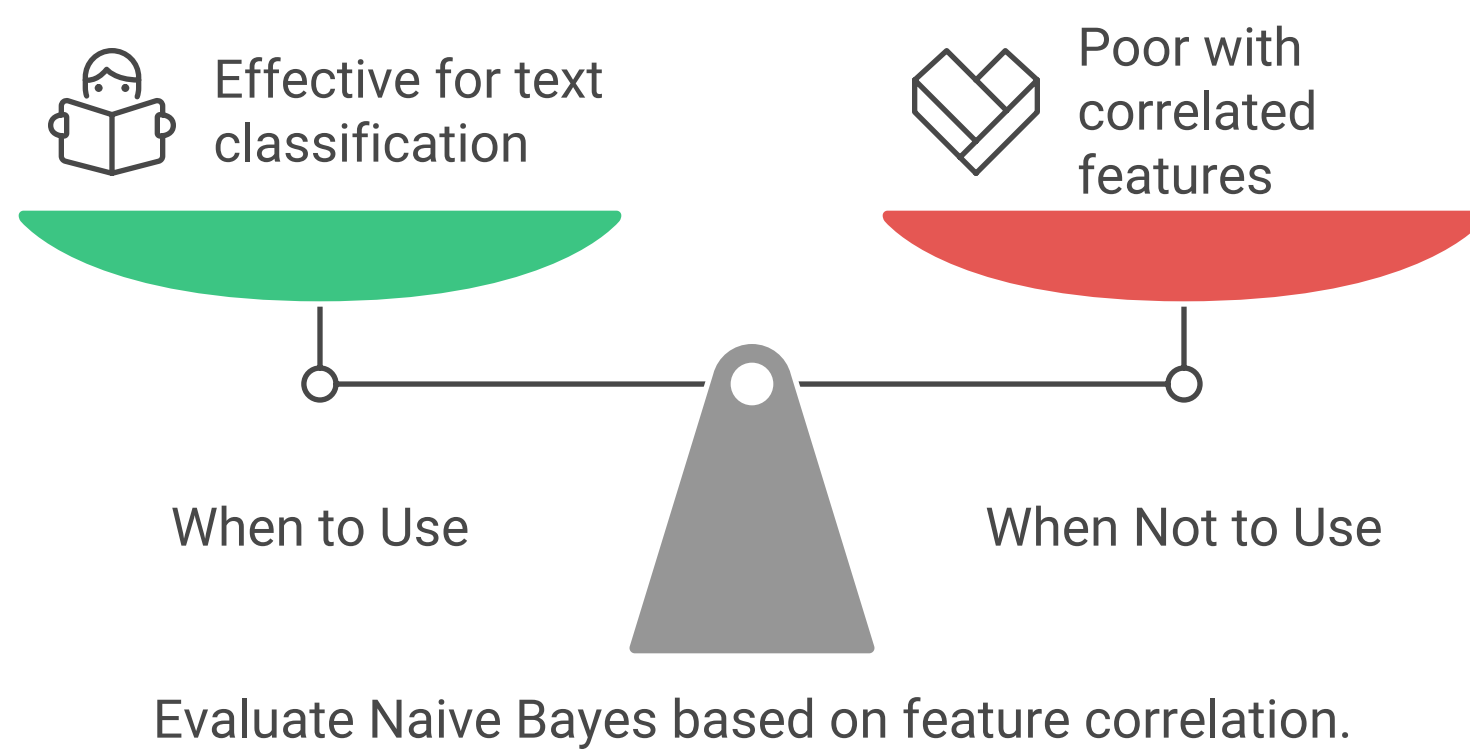
- **When to Use:** SVM regression is useful for predicting continuous outcomes, particularly when the relationship is complex and non-linear. It is robust to outliers and can handle high-dimensional data.
- **When Not to Use:** Similar to the classifier, SVM regression can be sensitive to the choice of kernel and parameters. It may not perform well with large datasets due to computational demands.



Balancing SVM Regression Strengths and Weaknesses

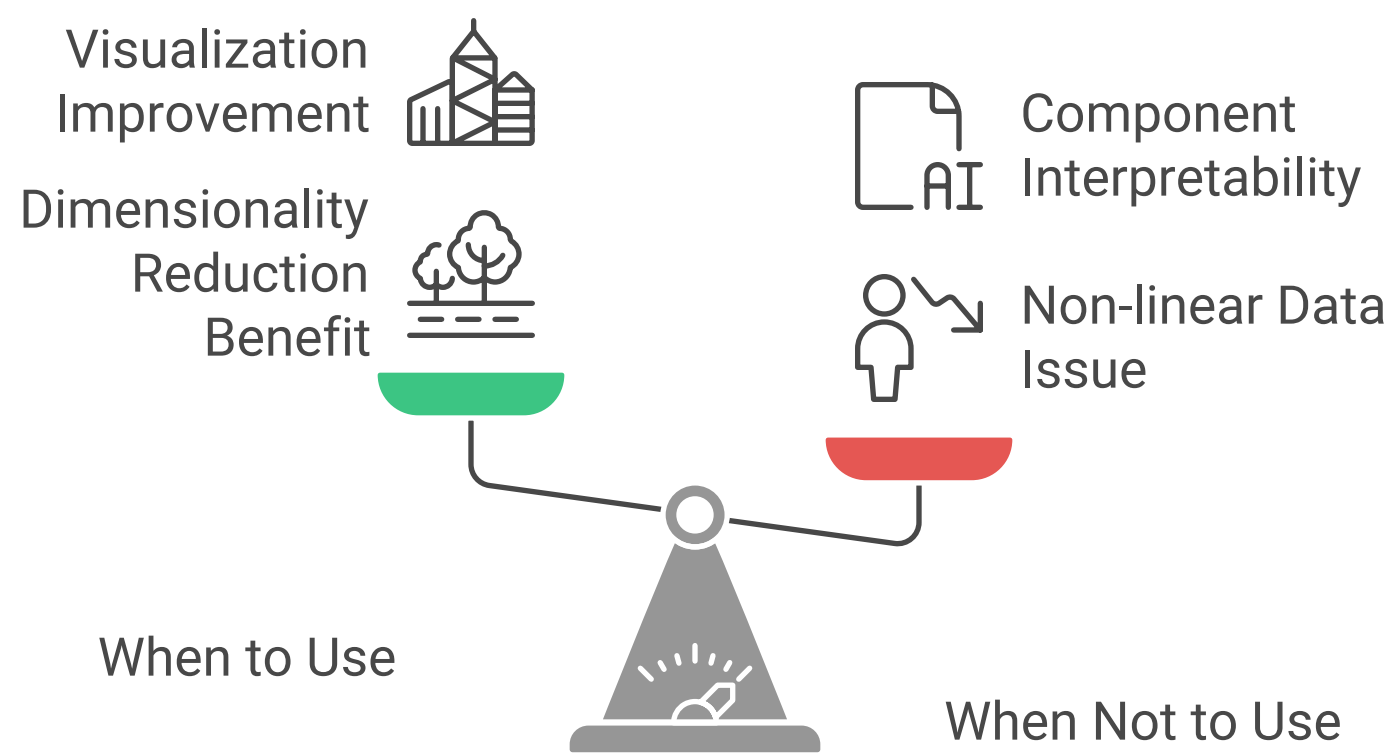
Naive Bayes

- **When to Use:** Naive Bayes is effective for text classification and problems with categorical input features. It works well when the independence assumption holds true.
- **When Not to Use:** It is not suitable for datasets where features are highly correlated, as the independence assumption may lead to poor performance.



Principal Component Analysis (PCA)

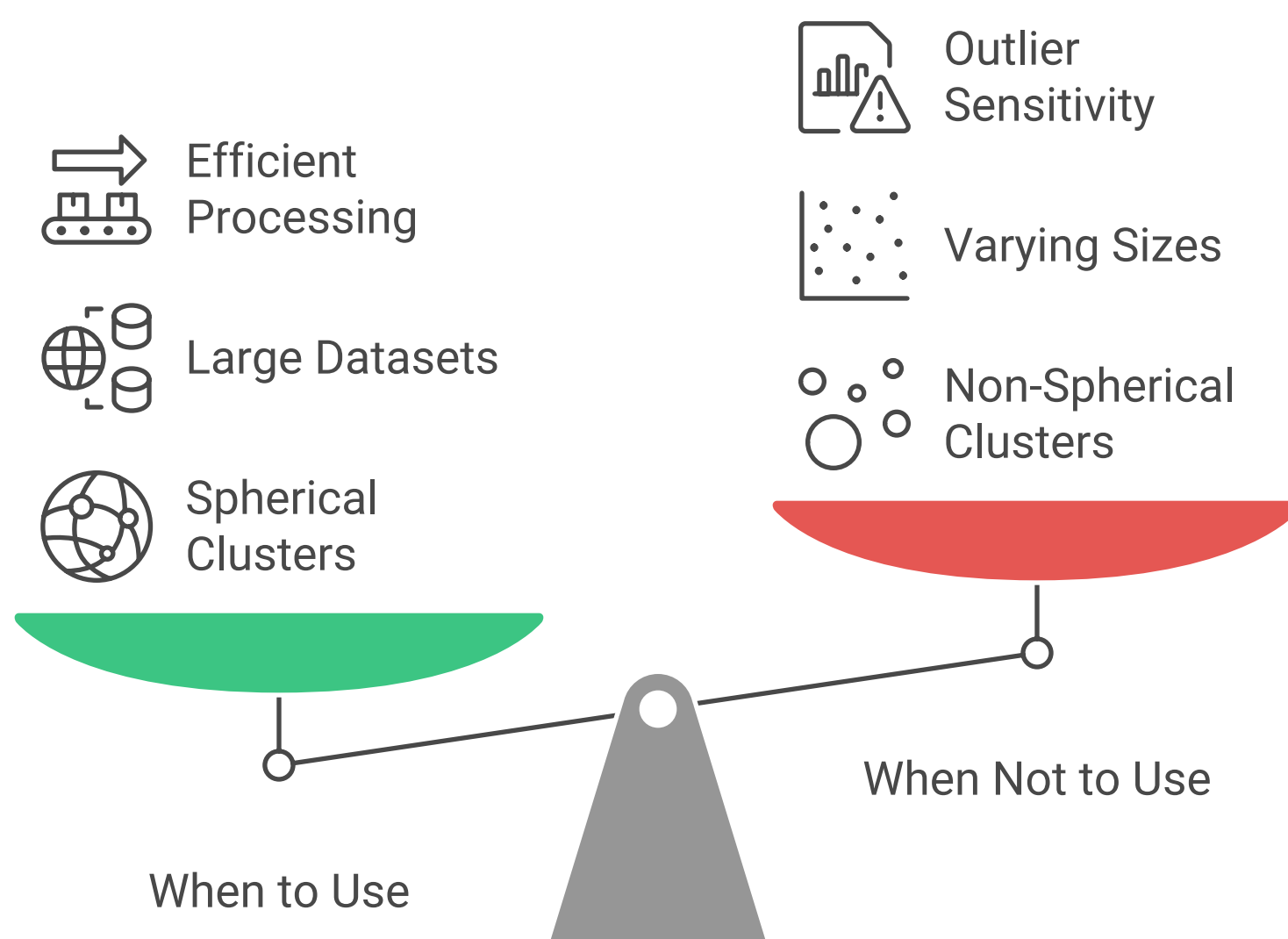
- **When to Use:** PCA is useful for dimensionality reduction, especially when dealing with high-dimensional data. It helps in visualizing data and improving model performance by reducing noise.
- **When Not to Use:** PCA may not be effective if the data is not linearly separable or when interpretability of components is crucial, as it transforms the original features into principal components.



Weighing PCA's Benefits and Limitations

K-Means

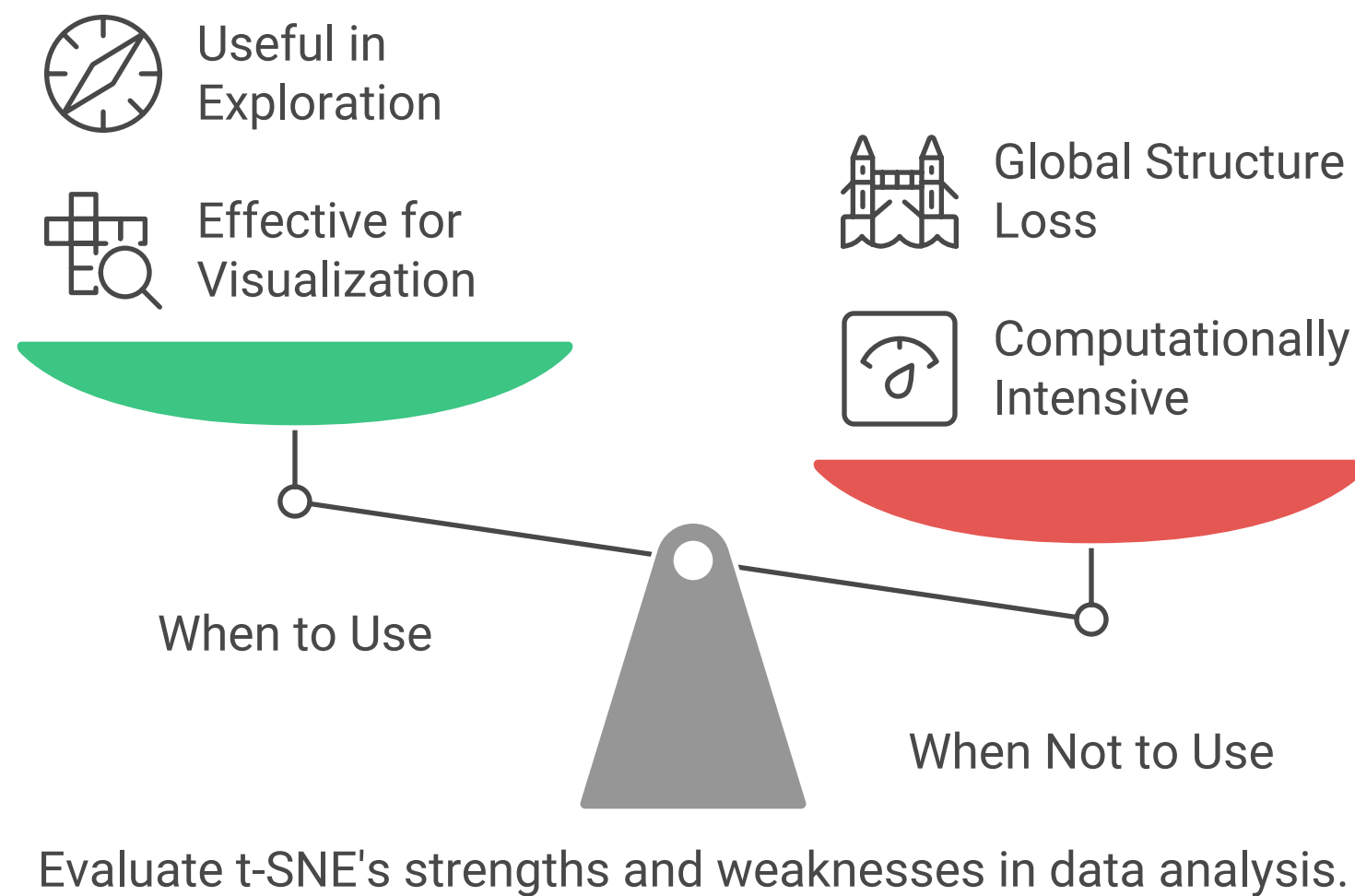
- **When to Use:** K-Means is suitable for clustering tasks where the number of clusters is known. It works well with spherical clusters and is efficient with large datasets.
- **When Not to Use:** It struggles with non-spherical clusters, varying cluster sizes, and outliers. The choice of the number of clusters [k] can also significantly affect results.



Evaluate K-Means suitability based on cluster characteristics.

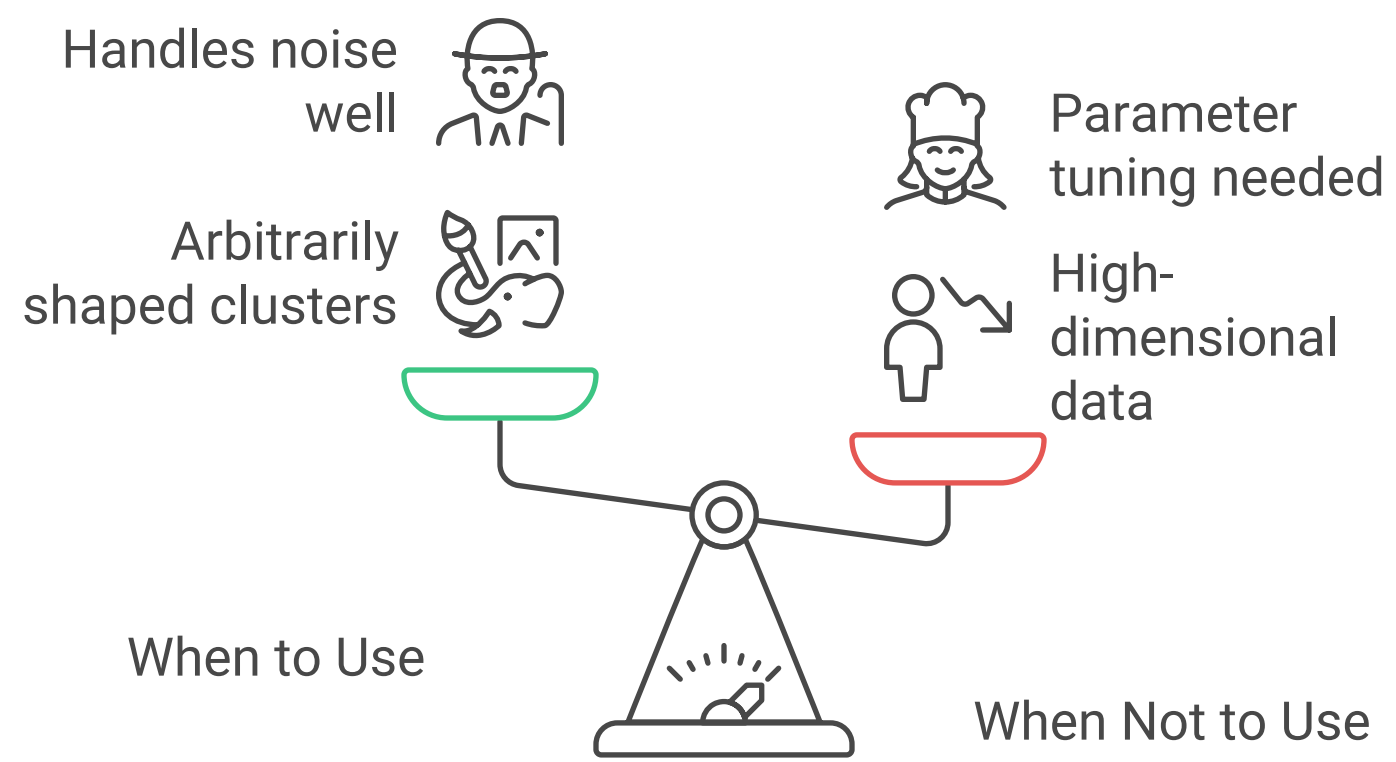
t-Distributed Stochastic Neighbor Embedding (t-SNE)

- **When to Use:** t-SNE is effective for visualizing high-dimensional data in two or three dimensions, particularly for exploratory data analysis.
- **When Not to Use:** It is computationally intensive and not suitable for large datasets. Additionally, it does not preserve global structures and is not ideal for clustering.



DBSCAN

- **When to Use:** DBSCAN is useful for clustering tasks, especially when dealing with noise and varying cluster densities. It can identify arbitrarily shaped clusters.
- **When Not to Use:** It may struggle with high-dimensional data and requires careful tuning of parameters (epsilon and min_samples). It is not effective when clusters are of varying densities.



Evaluate DBSCAN's strengths and limitations in clustering.

This overview serves as a guide to help practitioners choose the right machine learning algorithm based on their specific data characteristics and problem requirements.